MARRS: A Framework for multi-objective risk-aware route recommendation using Multitask-Transformer

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ABSTRACT

One of the most significant map services in navigation applications is route recommendation. However, most route recommendation systems only recommend trips based on time and distance, impacting quality-of-experience and route selection. This paper introduces a novel framework, namely MARRS, a multi-objective route recommendation system based on heterogeneous urban sensing open data (i.e., crime, accident, traffic flow, road network, meteorological, calendar event, and point of interest distributions). We introduce a wide, deep, and multitask-learning (WD-MTL) framework that uses a transformer to extract spatial, temporal, and semantic correlation for predicting crime, accident, and traffic flow of particular road segment. Later, for a particular source and destination, the adaptive epsilon constraint technique is used to optimize route satisfying multiple objective functions. The experimental results demonstrate the feasibility of figuring out the safest and efficient route selection.

CCS CONCEPTS

• **Computing methodologies** → *Multi-task learning*; Neural networks; • **Information systems** → **Spatial-temporal systems**; • **Applied computing** → *Transportation.*

KEYWORDS

recommendation system, multi-task learning, spatiotemporal data, graph neural network

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1 INTRODUCTION

Automatic route navigation systems provide route recommendation based on shortest distance and time-aware routes, which helps traveling in unfamiliar or complex routes. These navigation systems are heavily used by tourists or visitors during traveling. According to statistics travel, and tourism industry contributes around 10% of the country's GDP [21]. Therefore, the safety of tourists is more important to any country or regional government to raise the country's economic growth. However, the current recommendations lack in considering risk prone navigation routes, such as accident prone road segments, crime prone areas, etc. This paper addresses the issue of tourist safety and proposes a framework, namely MARRS, a risk-aware route recommendation system. The Figure 1 shows the green highlighted safer route where $\{O, D\}$ is the origin-destination pair and color represents the risk level (green-safer, red-risky, orange-moderate).



Figure 1: An scenario to risk aware recommendation

Existing literature on context-aware route recommendation systems consider single objective such as congestion [14], weather [12], crime [32], accidents [22], etc. Moreover, past few years, a multi-objective route recommendation system [23, 25] has come into existence, but based on coarse-grained analysis. On the other hand, the proposed framework provides fine-grained and multi-objective route recommendation systems. Another issue with the existing multi-objective route recommendation system is considering different objectives one after another in sequential manner, whereas we are framing the problem as minimizing multi-objective using the adaptive epsilon constraint technique. Moreover, when users set the weights based on their own judgement (e.g., significantly higher weight on crime than road accident), it may result in a route that is unbalanced (e.g., worst travel time and low risk), which is not what the users want. In the proposed work, despite the user entering extremely unbalanced weights, we attempt to prevent unbalanced routes in order to satisfy the user's customary wishes.

The challenges in developing the system include heterogeneous data sources, sparsity, complex spatio-temporal relation, and multi-objective optimization. To address the limitation of existing systems and resolve the challenges, we propose MARRS that unifies diverse spatial, temporal, and contextual data, inspired by the accessibility of open big data.

The following is a summary of the major contributions:

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- Inferring risk at the road level and providing risk-aware route recommendations using heterogeneous open data is a first-of-its-kind study, as far as we are aware. Large-scale urban sensing data are combined to allow for the precise prediction of fine-grained risk and the recommendation of the optimal risk-aware routes based on inferred road-level risk.
- We propose a novel wide, deep, and multitask-learning framework that collectively predict road accident, crime risk, and traffic flow for all road segments using heterogeneous open-data. We extract temporal, spatial, external, and recurrence features utilized by transformer based multitask-learning configuration to infer the risk on roads.
- Our approach is validated using datasets from the real world. The findings show that our framework performs better than other baseline methods in determining the level of risk on individual roads and efficiently creating risk-aware optimal routes.

The rest of the paper is organized as follows: Section 2 provides existing work in risk-aware route recommendation, and Section 3 explains important definitions to understand the problem. Proposed architecture given in Section 4 defines different components. Section 5 provide insights of data, baseline models, evaluation metrics and results. At last, Section 6 provides the conclusion and future work of the whole presented work.

2 BACKGROUND

We review the literature in this section in terms of two categories: context-aware route recommendation and spatial-temporal prediction systems. A brief overview is given as follows:

2.1 Context-aware route recommendation systems

In [24] the authors provided a survey about quality aware route navigation systems instead of focusing more on the shortest and fastest routes. In [32], the authors proposed road-level crime prediction system by using heterogeneous open data considering spatial, temporal, and recurrence features. Another system proposed by [31], analysed the crime from the spatio-temporal perspective and reported the effect of weather, POI, and taxi trips on crime. Authors' in [6] formulate the bi-objective problem that utilized crime data to provide short and safe route, whereas in our case, we are working on multi-objective problem. In [30], authors suggested methods for personalized navigation, determining a driver's context and then driving preferences for different situation using past driver trajectory data. Another work [18] proposed a multi-modal (bus, car, cycle) transportation recommendation system based on the contextual data using multi-task learning framework, whereas in our case, the main objective is to use multi-task learning for risk prediction.

2.2 Spatial temporal prediction systems

Since the spatial-temporal prediction using deep learning techniques have been studied from many years, such as traffic prediction [28], accident prediction [33], crime prediction [10], air quality prediction [27], and urban anomaly prediction [9]. In some of the recent works [9, 10, 20], prediction problems are course grained (region-based); on the other hand, we are doing fine grained (road segment-based) prediction. Fine grained risk prediction based on heterogeneous data provided by [32], but they predict crime only, whereas in MARRS, we deal with the multi-aspect risk-aware recommendation. Among these methods, the systems either made predictions for particular problems or real-life recommendations were not provided, making the systems less adaptable.

3 PRELIMINARY

In this section, we define terminology before describing the MARRS problem statement. We consider R urban roadways and J jobs/tasks (crime, accident, flow) with a T time window in particular.

3.1 Notation

Definition 3.1. **Road Network** A road network represented as a undirected graph $G = (N, E)$ is a geographical region consisting of a finite set of nodes $N = n1, n2, \ldots, n_z$ denotes the intersection points and a finite set of edges E where graph edge e_{ij} denotes a connectivity between node n_i and n_j. We have focused on undirected graphs because our approach primarily aims at pedestrians, who are more vulnerable to dangers in metropolitan environments. Users can go in both directions on roadway parts; however, we emphasize that this assumption is not required, and the proposed framework can be used for both directed and undirected graphs.

Definition 3.2. **Route** A route (R) on G is defined as a connected sequence of edges that connects a specific source, s, to a specific destination, d, with the first edge in the sequence starting in s and the last edge in the sequence finishing in d. We denote a route R by $R_{s,d}$ when we want to define the source and destination nodes. The length of a route R is equal to the sum of its edges' lengths, i.e., $l(R) = \sum_{e \in R} l(R)$.

Definition 3.3. **Crime Event** We used 5-tuples to represent a crime event that occurred at the coordinates (lat, long) of road n at time t with severity level s in crime type c as (c, n, t, lat, long, s). The number of crime events with severity level in a road n at time window t denoted by crime risk $C_r = \psi(n, t, s)$, where ψ is a function representing the summation of severity levels.

Definition 3.4. **Accident Event** We use 5-tuples to denote an accident occurrence that occurred in coordinate (lat, long) of road n at time t in accident type a as (a, n, t, lat, long, u). Number of accident events with severity level u in a road n at time window t denoted by accident risk $A_r = \varphi(n, t, u)$, where φ is a function represent the summation of severity levels.

Definition 3.5. **Traffic flow** Traffic flow in a road n at time window t is represented as 4-tuples (n,t,v,w), where v is the average traffic volume, and w is the average traffic speed. The traffic risk is $T_r = \gamma(n, t, v, w)$, where γ is a function of v, w in a road n at time window t.

3.2 Problem Statement

Prediction techniques for route recommendation typically rely on extracting spatio-temporal data, and at a high level the system accepts as an input the road network $G = (N, E)$, as well as a pair of

source-destination nodes, (S, D), and its purpose is to offer the user with a (set of) quick and safe paths between S and D. Accordingly, we divide the overall problem into two parts formulated as:

Problem 1. *Given the historical data with J tasks crime, accident, and traffic flow $J = (J_1, J_2, J_3)$ across roads $N = (n_1, n_2, \ldots, n_z)$ and k time step $T = (t_1, t_2, \ldots, t_k)$ of a city, the objective is to learn a risk aware predictive function that infer crime (C_r), accident (A_r), and traffic risk (T_r) at each road segment n in f future time steps. Formally, it can be represented as*

$$y^{(k+f)} = \Lambda(\mathcal{G}, C_r, \mathcal{A}_r, \mathcal{T}_r, \mathcal{E}_x); \quad (1)$$

where mapping function $\Lambda(.)$, we want to learn that have road network ($\mathcal{G}$), crime risk (C_r), accident risk ($\mathcal{A}_r$), Traffic risk ($\mathcal{T}_r$), and external data ($\mathcal{E}_x$) such as weather, POI, anomaly, calendar data, etc. as arguments. The output is $y(k + f)$ which represents the prediction of all risk events in all roads in f future time steps.

Problem 2. *If R denotes the collection of all potential routes from S to D, our goal is to find a small subset of paths $P \subseteq R$ such that for any path $R \in P$, R is neither dominated by nor equal to any other path R' in P. Given two paths R and R' and assuming that their lengths and risks can be computed using functions $\lambda(.)$ and $\Lambda(.)$, respectively, which are to be minimized, we say that R dominates R' if $(\lambda(R)) \leq (\lambda(R'))$ and $\Lambda(R) \leq \Lambda(R')$ and at least one of the two inequalities is strict, that is, either $(\lambda(R)) < (\lambda(R'))$ and $\Lambda(R) < \Lambda(R')$.*

The problem with this high-level problem formulation is that the shortest path (in terms of length) is not always the smaller risk (in terms of crime and accident), and vice versa. As a result, the problem we're trying to solve is a multi-objective optimization problem in which the goal is to reduce the given path's length as well as its risk.

4 OVERALL FRAMEWORK OF MARRS

The framework shown in Figure 2 has different layers such as datasets, feature extraction, deep learning, risk-aware recommendation, quality analysis, etc. We begin by gathering a vast amount of important heterogeneous data, such as crime, accident, traffic flow, demographic, taxi mobility, POIs, and meteorological data. We next extract collection of features, such as temporal, spatial, external, and recurrent crime features from the data. We incorporate temporal and spatial information into a multi-task learning model in the risk prediction phase to generate spatio-temporal estimation of risk occurrences on highways. We can present a set of risk-aware route recommendation ideas based on these inferred dangers on roadways, given the source and destination locations, taking into account both route distance and possible risk. In the following sections, we go over the framework in more detail.

4.1 Dataset Collection

With the increase in urbanization, the online system has also come into existence in developed countries. Therefore, to conduct the experiments, we use different real-world datasets from NewYork City [1] to validate our proposed framework and separately perform experiments to predict crime, accidents, and traffic during the period from 01/01/2018 to 31/12/2018. The distribution of accident, crime,

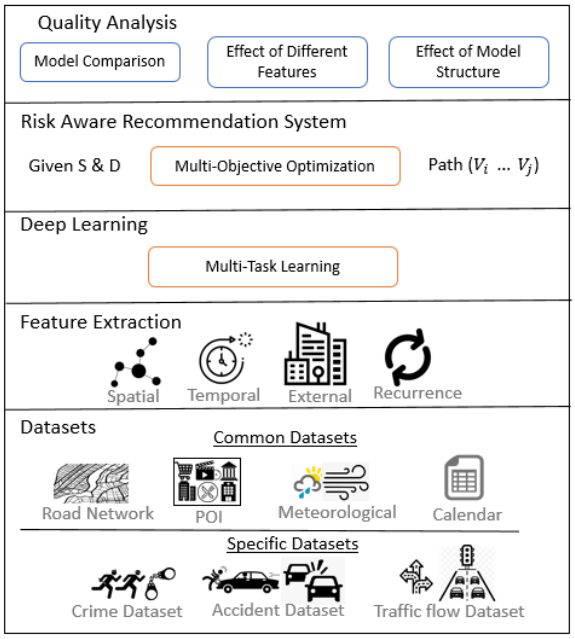


Figure 2: Framework for recommendation system

and POI is shown in Figure 3, where darker color represents more number of accidents/crime.

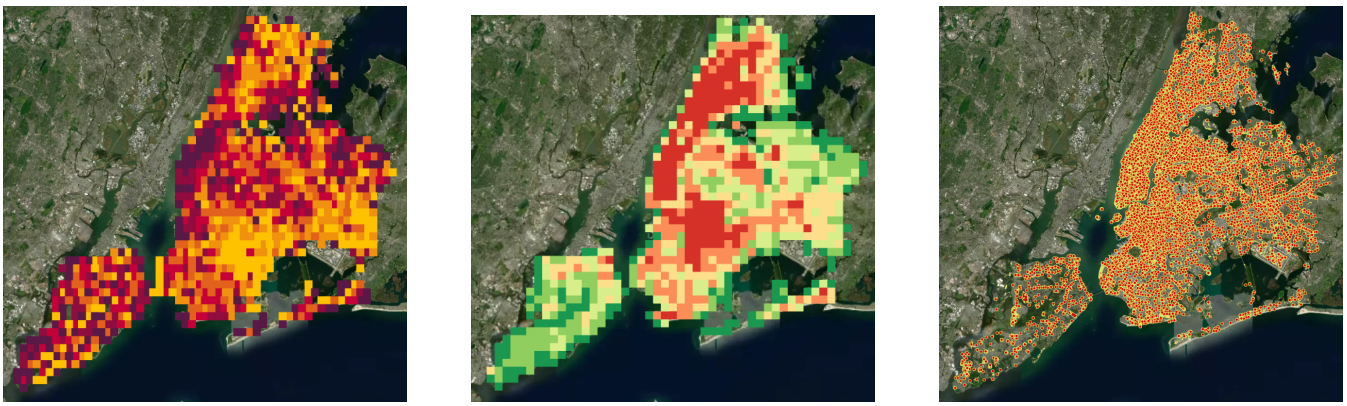
- *Traffic accident Data*: The New York Police Department's statistics on motor vehicle collision is included in the dataset (NYPD) [2]. The timestamp and coordinate (latitude and longitude) information of the accompanying traffic accident is recorded with each accident.
- *Crime Event Data*: The crime data provided by NYPD record all valid crime events that residents reported to NYPD. The dataset contains all valid crime types, such as larceny, assault, burglary, felony, misdemeanor, violation, etc., with location and timestamp.
- *Anomaly Data*: The city of New York has a 311 emergency service platform [3] that lets residents file complaints by phone call, text message, or mobile app. We extracted four columns from the complaint format, i.e., time stamp, complaint type, latitude, and longitude, and conducted studies on two categories of anomaly (complaints): blocked driveways and illegal parking, due to the potential impact on the dynamics of the road.
- *Vehicle Speed Data*: The dataset [4] contains real-time traffic data from places throughout the five boroughs where NYCDOT collects sensor feeds, primarily on major arterials and highways. The Traffic Management Center (TMC) of the New York City Department of Transportation (NYCDOT) maintains a map of traffic speed detectors located across the

[1] https://opendata.cityofnewyork.us/

[2] https://data.cityofnewyork.us/Public-Safety/Motor-Vehicle-Collisions-Crashes/h9gi-nx95

[3] https://portal.311.nyc.gov/

[4] https://data.cityofnewyork.us/Transportation/Real-Time-Traffic-Speed-Data/qkm5-nuaq



(a) Accident Distribution (b) Crime Distribution (c) Point of Interest (POI)

Figure 3: Distribution of accident, anomaly and POI in NYC city

city. The speed detectors are owned by a variety of city and state authorities.

- *NYC-POI*: Point of interest contains data about different categories like education, financial, transportation, healthcare, entertainment, etc., with geo-coordinates.
- *NYC-Road Network*: The primary transportation system infrastructure is the road network, and we collect the road network dataset of NewYork city from open street map (OSM) using python package OSMnx [1].
- *NYC-Meteorological Data*: We crawl weather data of New York city from WorldWeatherOnline [5] that include different weather types like wind, cloud, temperature, humidity, pressure, visibility, heat index, etc. with the interval of 1 hour.
- *NYC-Calender*: The calendar data like weekday, weekend, holiday obtained from Holiday library [6].

4.2 Feature extraction

We extract spatial, temporal, external, and recurrence features from the heterogeneous datasets. The details about each are given as follows:

4.2.1 Temporal feature. Historical data of crime, accident, and traffic risks are multivariate timeseries data along with spatial data. Human mobility might influence the occurrence of risks. For example, a rise in the number of persons in the public may raise the likelihood of committing a crime, such as larceny. Due to a lack of data, we've decided to use cab trip data as a measure of human mobility.

4.2.2 Spatial feature. The distribution of POIs on the road may signify the usefulness of that road. For example, more entertainment facilities, such as nightclubs, on one street, may draw more drunken brawls. The distribution of POIs across ten categories is calculated. As is generally known, a given location's socioeconomic circumstances impact risk rates; therefore, some of the work also incorporated demographic features.

[5]https://www.worldweatheronline.com/
[6]https://pypi.org/project/holidays/

4.2.3 External feature. Date and time, weather conditions, and air temperature are all temporal contextual elements linked to risks, according to previous studies. Risks on the roads appear to be time-dependent. For example, there are very few people outside during business hours, resulting in a very low risk of risk commitment. It has been discovered that risk occurrences differ between weekdays and weekends/holidays, as well as between different temporal duration. As a result, we divide the 24 hours of different day types across various temporal periods. Prior research has shown that risk incidences can vary depending on meteorological circumstances. During hazy days, for example, close-range visibility might be obscured, preventing potential risk commitments. Many individuals prefer to go out on clear days, which increases the risk of some risky situations, such as petty larceny. As a result, we take meteorological data from the year 2018 and average daily risk occurrences under four different weather circumstances. On cold and hot days, more individuals prefer to stay indoors, which reduces the likelihood of risk. On hot days with a large number of people outside, several risks, such as illegal parking and larceny, are more likely to occur.

4.2.4 Recurrence feature. The occurrences of risk events would also be influenced by surrounding previous dangers, i.e., the near-repeat phenomenon. This phenomenon suggests that risk occurrences may recur in close vicinity, both in terms of time and distance. According to this idea, risk occurrences in one location are influenced by historical local risk occurrences and historical risk occurrences in neighbouring regions. As a result, the risk events that occurred would become a predictor for future activity. The recurrence risk feature is defined as a characteristic that represents near-repeat influential elements for future hazards.

4.3 Deep learning architecture

For forecasting criminal risk, accident risk, and traffic risk, we introduce WD-MTL (wide and deep multi-task learning) inspired by architecture given by authors' [18], with the difference in task prediction and shared layer transformer. As shown in the Figure 4, the wide component, the deep component, and the task-specific component constitute the WD-MTL. The deep component is shared

by all tasks, and each task has its own wide component and task-specific component. The wide component and the deep component of each task are identical to the basic model. There are two types of characteristics in each task: generic features and task-specific features. Task-specific features such as crime, accident, and traffic data vary in different tasks; however, general features such as temporal and meteorological features are the same for all activities. We used multi-task learning (MTL) since it is used to address scenarios in which multiple tasks with some commonality are offered. Due to the overlapping properties of the tasks, such an optimization technique can increase generalization and accuracy for task-specific models compared to training the models independently.

4.3.1 Multi-task learning. To capture personalized information, we employ the MTL model, which has been utilized in numerous recommendation systems. Instead of training separate neural networks for each task, MTL may learn different tasks in a single model, which is more effective in terms of data as well as memory and inference speed. In our case, a risk prediction is one task, i.e., J_i where $i = 1, 2, 3$ are three different tasks. For example, crime risk prediction is one task, and accident prediction is another. For multi-task learning, we adopt a shared bottom approach. The MTL shared bottom approach is formalized as follows:

$$y_i = \vartheta^i(\varrho(x)) \tag{2}$$

Here, y_i is the output parameter for each risk prediction task. Given i^{th} task, the model comprised of shared bottom network as function ϱ, and n tower networks ϑ^i, where $i = 1, 2, \ldots, n$ for each task respectively. Any neural network can be incorporated into multi-task learning configuration. Here, we use a spatial-temporal transformer network to explain the approach.

4.3.2 Spatial Block. The geographic area is separated into regions for fine-grained analysis or complexity reduction in various applications, such as traffic prediction, urban anomaly prediction, and trip planning, which gives semantic information across spatial axes. The relationship between geographical regions based on the connection of road networks is represented by the region graph $G = \{N, E\}$. Road connections tend to bring together areas that share characteristics. If regions are connected, inflow and outflow of traffic are more likely to influence problems with transport risk prediction. Using a spatial transformer, this spatial relationship is captured. The positional embedding layer, static and dynamic layers, and gate fusion layer make up the spatial transformer [29]. Each node in the positional embedding layer contains spatial temporal position information, such as topology, connection, and time steps. We make use of the position embedding suggested by authors' [4] that takes the idea of Laplacian eigenvectors. Intuitively, urban area signals can be divided into stationary (such as the road network and buildings) and dynamic (e.g., weather, accidents, in-out-flow, etc.). As a result, the framework makes use of both static and dynamic layers, and the results of these layers are then combined to produce the final output. The graph convolution [15] technique is utilised in the static layer, where each node collects the characteristics of neighbouring nodes based on a predefined graph, with the outcome represented by $\hat{Y}^{sl}$. The hidden time-evolving patterns are captured by the dynamic layer. The projected output of the embedded positional layer $\hat{X}^{sp}$ is a three-dimensional latent space that is represented by

$$Q^{sp} = X^{sp}U_q^{sp} \tag{3}$$

$$K^{sp} = X^{sp}U_k^{sp} \tag{4}$$

$$V^{sp} = X^{sp}U_v^{sp} \tag{5}$$

Here, the weight matrices for the query Q^{sp}, key K^{sp}, and value V^{sp} are represented as $U_q^{sp}, U_k^{sp}, U_v^{sp} \in \mathbb{R}^{d_f \times d}$. The score calculation for determining self-attention, which requires the dot product of the Q^{sp} and K^{sp} vectors, is the next step, similar to the transformer [26]. The result is normalised and softmaxed, and given as

$$S^{sp} = softmax\Big(\frac{Q^{sp} \cdot K^{sp}}{\sqrt{d}}\Big) \tag{6}$$

The next step is to multiply V^{sp} and S^{sp} to ignore unnecessary patterns, using the formula $M^{sp} = V^{sp} \cdot S^{sp}$. The multi-headed attention mechanism captures various spatial dependencies, and the output h^{l+1} is then transmitted to the feed forward neural network by leveraging the residual connection and norm layer.

$$\hat{Y}^{sh} = ReLU\Big(U^l, (N_{norm}(h^l + h^{l+1}))\Big) \tag{7}$$

The gating layer is used to fuse the output of static and dynamic layer represented as

$$\Gamma = \sigma\Big(\Upsilon_{sl}(\hat{Y}^{sl}) + \Upsilon_{sh}(\hat{Y}^{sh})\Big) \tag{8}$$

Here, the linear projections are Υ_{sl} and Υ_{sh}.

$$\hat{Y}^{sp} = \Gamma(\hat{Y}^{sl}) + (1 - \Gamma)(\hat{Y}^{sh}) \tag{9}$$

Here, the k time steps of $\hat{Y}^{sp}$ are calculated and fed as an input into the temporal block.

4.3.3 Temporal Block. Long-range dependencies are effectively maintained via temporal transformers [26]. The temporal transformer is more easily scaled to extended sequences with concurrent processing of long-range dependencies than RNNs and its variants. Similar to the spatial transformer, the temporal transformer has the formula $\hat{X}^{tp} = \zeta(X^{tp}, E^{tp})$. To simulate temporal interdependence, the self-attention mechanism is also used. The temporal transformer receives a temporal sequence $\hat{X}^{tp}$ with a slide window of length k as its input. Temporal dependencies, which include the query Q^{tp}, the key K^{tp}, and the value V^{tp}, are dynamically calculated in high-dimensional latent subspaces, much like the spatial transformer.

$$Q^{tp} = X^{tp}U_q^{tp} \tag{10}$$

$$K^{tp} = X^{tp}U_k^{tp} \tag{11}$$

$$V^{tp} = X^{tp}U_v^{tp} \tag{12}$$

Where U_q, U_k, U_v are learned linear mappings.

$$T^{tp} = softmax\Big(\frac{Q^{tp} \cdot K^{tp}}{\sqrt{d}}\Big) \tag{13}$$

The next step is to multiply V^{tp} and S^{tp} in order to ignore patterns that are unimportant, using the formula $M^{tp} = V^{tp} \cdot S^{tp}$. Through a multi-headed attention mechanism, different temporal dependencies are collected. The output, h^{l+1}, is then transferred to a feed-forward neural network by using a residual connection, and

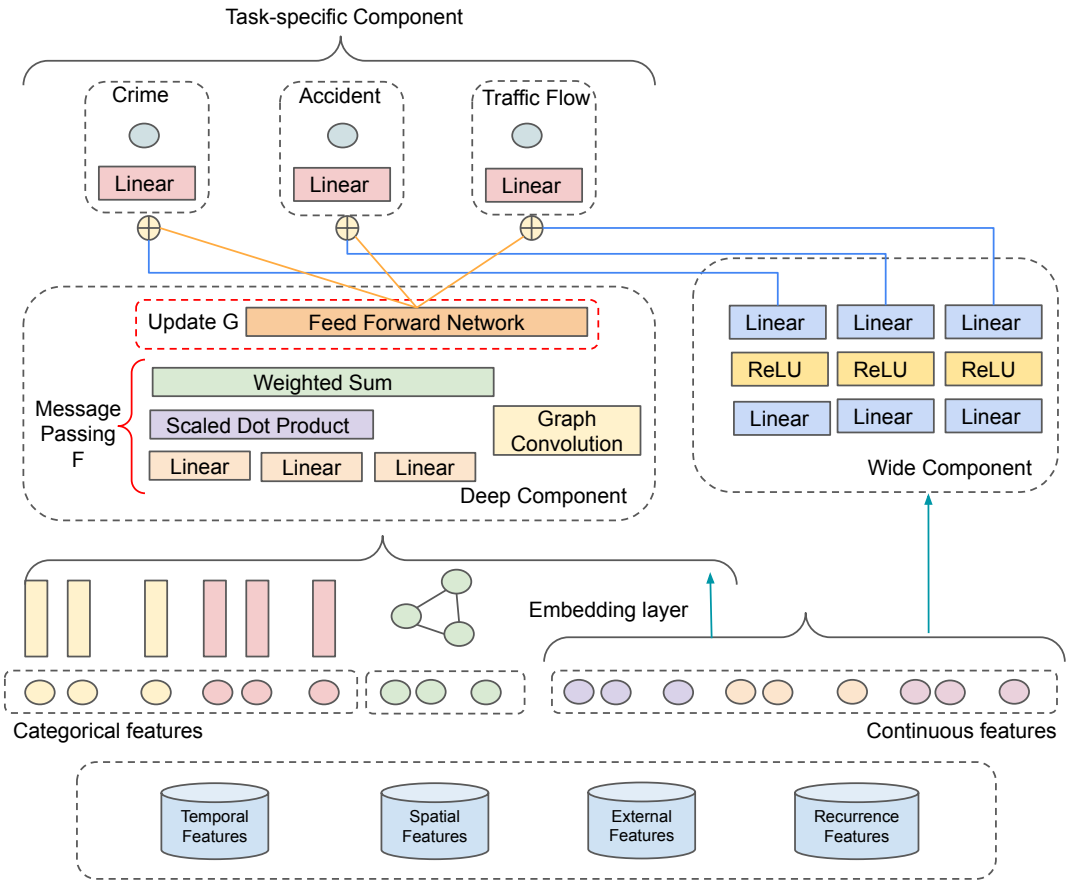


Figure 4: Graphical representation of proposed architecture

the norm layer represents the output as $\hat{Y}^{tp}$, which is then passed to the task-specific component.

4.3.4 Prediction Block. In the prediction phase of proposed MARRS architecture, we utilize the Multi-Layer Perceptron (MLP) as a regressor to generate the occurrence of risks in different segments. The MLP is able to capture non-linear relationship in the data and anomaly occurrence probabilities represented as the class of neural network i.e., $y^k = \phi(wx + b)$, where w and b are learnable parameters and ϕ is ReLU activation function. x represents the fused output from deep (spatial and temporal block) and wide component. The loss function is defined in terms of Huber loss [11]. The squared error loss is more susceptible to outliers than the Huber loss.

4.4 Risk aware recommendation system

We used the same concept as in [6] for finding the safest path, except we are working on multi-objective optimization, and they proposed a bi-objective optimization solution. The proposed approach aims to efficiently explore the space of non-dominated pathways from s to t given a source-destination pair (s, d). In our case, we shall utilize the Dijkstra shortest route algorithm, represented as $Dijkstra(G, \Delta())$ shows that the Dijkstra shortest-path algorithm was applied to the input graph G using edge weights defined by the function $\Delta()$. Overall, the $Dijkstra(G, \Delta())$ produces a route R in G that minimises $\sum_{e \in R} \Delta(e)$. Any single source shortest path algorithm can take the place of the Dijkstra algorithm. Here, the function Δ is the adaptive epsilon constraint function [17] which optimizes three objective i.e., road accident, crime, and traffic flow. We employ the product of a normalized objective's value and its weight (given by a user) as an input (rather than the objective's value) to account for user preference.

5 EVALUATION

The datasets from New York City are used in experiments to evaluate the usefulness of the proposed framework. We first go over the experimental setup in this part, including parameter settings, baselines, and performance metrics, then validate our proposed framework from many perspectives with the goal of providing answers to the following:

- **Q1** How well does our framework predict risks when compared to baseline techniques?
- **Q2** What effects do different component combinations have on performance when used in a joint architecture?
- **Q3** How does the framework's overall performance change as a result of the context information?

5.1 Experimental setup

5.1.1 Parameter settings. All of the experiments were carried out on Google Colab using a Tesla P100-PCIE-16GB GPU and an Intel Xeon 2.20 GHz CPU. The Colab environment allotted 12 GB RAM and 34 GB of hard drive space for memory. Using the PyTorch toolkit, we implement the model in Python. Each categorical feature is converted into a 32-dimensional embedding vector in the deep component's embedding layer before being concatenated with all of the continuous features as the input vector. Two fully connected layers receive the input vector afterwards. A ReLU activation function and 300 hidden units constitute each fully connected layer. The multi-task component receives its input features from the last fully connected layer's output. Each task-specific network learns from shared bottom layers in a multitask environment; in our approach, we used a transformer. We are taking into account three different risks related to crime, accidents, and traffic (i.e., three tower networks are used), which comprised four linear layers, each of size 200 is used in task-specific layer. Three linear layers, each 300 units wide, constitute a wide component. With a learning rate of $1e-3$, we train the model using the Adam optimizer. The experiment's batch size is set to 512, and the regularization parameter *lambda* is set to 0.01.

5.1.2 Baselines. We compare MARRS with different baselines, i.e., classical predictive analytic models (i.e., SVR, LR), conventional forecasting methods (i.e., ARIMA), and variants of recurrent neural networks (i.e., ST-RNN, LSTM, GRU, ARM).

- Support Vector Regression (SVR) [2]: Supervised machine learning based method for regression based on kernel methods which are characterized by support vectors and margin.
- Logistic Regression (LR) [8]: This method is a statistical model that predicts each region's anomaly occurrence based on historical data.
- Auto-Regressive Integrated Moving Average (ARIMA) [13]: Time-series prediction method based on the assumption that current and past time-series values have a linear relationship.
- Spatial-Temporal Recurrent Neural Network (ST-RNN) [19]: Spatial and temporal relation capture with the help of recurrent neural network.
- Long Short Term Memory (LSTM [7]): Improved version of basic RNN to remove the vanishing gradient problem and works well with time-series data.
- Gated Recurrent Unit Neural Network (GRU [3]): This is based on a gating mechanism to predict the event in future time step.
- Attention based Recurrent Neural Network (ARM) [5]: This method use attention mechanism on recurrent neural network to predict spatial and temporal mobility.

5.1.3 Performance measures. Performance measures are used to evaluate the model quantitatively; we use performance measures, namely, mean absolute error (MAE), mean relative error (MRE), and root mean square error (RMSE), as metrics for evaluation as they are popular metrics for regression.

$$MAE = \frac{1}{K}\sum_{m=1}^{K}|y_m - \hat{y}_m| \tag{14}$$

Table 1: Comparison with different baselines in predicting average risks

Methods	MAE	MRE	RMSE
SVR	0.87	0.63	1.45
LR	0.89	0.66	1.55
ST-RNN	0.89	0.81	1.48
LSTM	0.88	0.71	1.46
GRU	0.85	0.70	1.44
ARM	0.87	0.77	1.46
MARRS	**0.75**	**0.55**	**1.37**

Here, MAE denotes the mean of each prediction error's absolute values across all instances (K). Prediction error $|y_m - \hat{y}_m|$ is the difference of actual risk score (y_m) and predicted risk ($\hat{y}_m$) score of a model.

$$MRE = \frac{1}{K}\sum_{m=1}^{K}\frac{|y_m - \hat{y}_m|}{y_m} \tag{15}$$

MRE is another variant of MAE with prediction error $|y_m - \hat{y}_m|$ is normalized.

$$RMSE = \frac{1}{K}[\sum_{m=1}^{K}(y_m - \hat{y}_m)^2]^{\frac{1}{2}} \tag{16}$$

RMSE is square root of MAE and due to the square root, RMSE does not penalise errors as severely as MSE does. The model predictions are better if MAE, MRE, and RMSE values are close to zero.

5.2 Performance validation

5.2.1 RQ1: Comprison with different baselines. Table 1 displays the results of urban anomaly prediction in terms of MAE, MRE, and RMSE in order to validate the performance of MARRS. The evaluation results show that MARRS outperforms existing baseline approaches in the task of predicting risk score. This clarifies the advantage of taking a spatio-temporal transformer into account. Second, approaches based on neural networks perform better than methods based on standard machine learning because neural networks are better at learning hidden and non-linear correlations. Last but not least, predicting risks with high accuracy requires geographical embedding, temporal, external, and recurrence information in addition to time-series data.

5.2.2 RQ2: Comprison with different components. We evaluate variants by doing ablation studies in addition to comparing MARRS with baseline techniques to understand the effects of spatial, temporal, and wide blocks. An ablation study determines how a specific component affects the functionality of the framework. From all the variants, it can be seen that the spatial and temporal elements are crucial for accurate prediction, as follows:
Effect of spatial block (S_t): A simplified version of the framework that does not consider spatial data for evaluation. As shown in Figure 5, the framework shows RMSE of S_t is 2.35, which is comparatively more than the actual RMSE value. It can be inferred from the result that spatial information provides additional context regarding prediction.
Effect of temporal block (T_t): Another framework variant does not

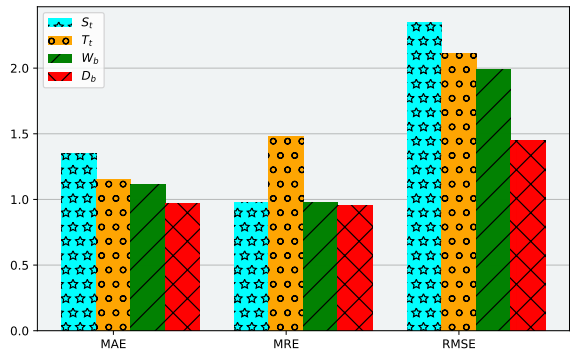


Figure 5: Component based comparison

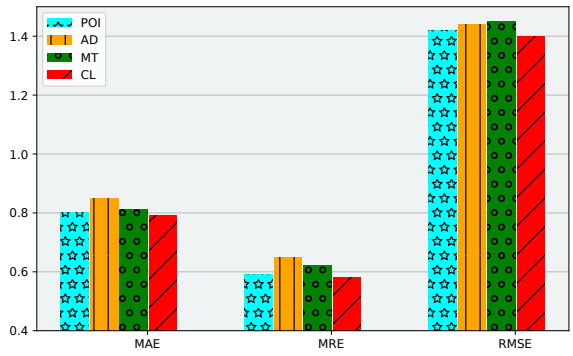


Figure 6: Context based comparison

cover the temporal aspect of the risk prediction problem. As observed from the results shown in Figure 5 that T_t has RMSE of 2.11, which is more than the overall framework. This effect is raised because risk changes with respect to time, and the temporal component plays an important role in prediction.

Effect of wide block (W_b): Wide block helps to improve the performance of risk prediction correctly, and the results shown in Figure 5 validate this statement. W_b has RMSE of 1.99, a significant reduction in the accuracy of the proposed framework. Therefore, adding a wide block helps in predicting urban risk more accurately.

Effect of deep block (W_b): Deep block help to improve the performance of risk prediction correctly, and the results are shown in Figure 5 to validate this statement. W_b has RMSE of 1.45, a significant reduction in the accuracy of the proposed framework.

5.2.3 RQ3: Context information effect. For better understanding, we also evaluate the framework from another perspective, such as how various contextual factors (POI, weather, and the calendar) affect the result. To increase the accuracy of predictions, we incorporate context information into our system. The model is given additional information by the POI (POI), anomaly data (AD), meteorological data (MT), and calendar (CL) context. The impact of these on the model's performance is something we are curious about. Figure 6 presents the findings, where context-PI implies the absence of specific context data. According to observation, point-of-interest data significantly affect all contexts.

5.3 Discussion

Our paper provides a way of predicting risk in urban areas. Further, it helps tourists and the general public take safe routes to mitigate threats and losses. However, MARRS has a few limitations that we want to discuss. First, the datasets collected from NYC OpenData services inherit all the issues of crowdsourcing data collection like data bias [16]. For example, all the accidents and crimes are not registered in the portal, and all registered data is not validated. Second, the model may not be adaptable to other cities as spatial-temporal data distribution will vary. At last, we want to conclude that despite the above limitations, our model performs better over baselines.

6 CONCLUSION

Using large-scale heterogeneous urban data, we investigated the challenge of risk inference at the road level in this research. We use the spatio-temporal aspects retrieved from heterogeneous sensing data, such as mobility, infrastructural, meteorological, demographic, and other types of data. The spatio-temporal features are then fed into a WD-MTL model for predicting accident, crime, and traffic risks using WD-MTL architecture. Then we structured the problem as a multi-objective optimization problem, with the goal of identifying a small range of nondominated pathways with varying distance-to-safety tradeoffs. Our findings demonstrated the efficacy and efficiency of our proposed framework.

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REFERENCES

[1] Geoff Boeing. 2017. OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems* 65 (2017), 126–139.
[2] Chih-Chung Chang and Chih-Jen Lin. 2011. LIBSVM: A library for support vector machines. *ACM transactions on intelligent systems and technology (TIST)* 2, 3 (2011), 1–27.
[3] Junyoung Chung, Caglar Gulcehre, KyungHyun Cho, and Yoshua Bengio. 2014. Empirical evaluation of gated recurrent neural networks on sequence modeling. *arXiv preprint arXiv:1412.3555* (2014).
[4] Vijay Prakash Dwivedi and Xavier Bresson. 2020. A Generalization of Transformer Networks to Graphs. *arXiv preprint arXiv:2012.09699* (2020).
[5] Jie Feng, Yong Li, Chao Zhang, Funing Sun, Fanchao Meng, Ang Guo, and Depeng Jin. 2018. Deepmove: Predicting human mobility with attentional recurrent networks. In *Proceedings of the 2018 world wide web conference*. 1459–1468.
[6] Esther Galbrun, Konstantinos Pelechrinis, and Evimaria Terzi. 2016. Urban navigation beyond shortest route: The case of safe paths. *Information Systems* 57 (2016), 160–171.
[7] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural computation* 9, 8 (1997), 1735–1780.
[8] David W.. Hosmer and Stanley Lemeshow. 2000. *Applied logistic regression*. Wiley New York.
[9] Chao Huang, Chuxu Zhang, Peng Dai, and Liefeng Bo. 2020. Cross-Interaction Hierarchical Attention Networks for Urban Anomaly Prediction.. In *IJCAI*. 4359–4365.
[10] Chao Huang, Chuxu Zhang, Jiashu Zhao, Xian Wu, Dawei Yin, and Nitesh Chawla. 2019. Mist: A multiview and multimodal spatial-temporal learning framework for citywide abnormal event forecasting. In *The World Wide Web Conference*. 717–728.

[11] Peter J Huber. 1992. Robust estimation of a location parameter. In *Breakthroughs in statistics*. Springer, 492–518.
[12] Muhammed Mas-ud Hussain, Besim Avci, Goce Trajcevski, and Peter Scheuermann. 2016. Incorporating weather updates for public transportation users of recommendation systems. In *2016 17th IEEE International Conference on Mobile Data Management (MDM)*, Vol. 1. IEEE, 333–336.
[13] Mehdi Khashei, Mehdi Bijari, and Gholam Ali Raissi Ardali. 2009. Improvement of auto-regressive integrated moving average models using fuzzy logic and artificial neural networks (ANNs). *Neurocomputing* 72, 4-6 (2009), 956–967.
[14] Andrew Kinai, Reginald E Bryant, Aisha Walcott-Bryant, Eric Mibuari, Komminist Weldemariam, and Osamuyimen Stewart. 2014. Twende-twende: a mobile application for traffic congestion awareness and routing. In *Proceedings of the 1st International Conference on Mobile Software Engineering and Systems*. 93–98.
[15] Thomas N Kipf and Max Welling. 2016. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907* (2016).
[16] Constantine E Kontokosta and Boyeong Hong. 2021. Bias in smart city governance: How socio-spatial disparities in 311 complaint behavior impact the fairness of data-driven decisions. *Sustainable Cities and Society* 64 (2021), 102503.
[17] Marco Laumanns, Lothar Thiele, and Eckart Zitzler. 2006. An efficient, adaptive parameter variation scheme for metaheuristics based on the epsilon-constraint method. *European Journal of Operational Research* 169, 3 (2006), 932–942.
[18] Hao Liu, Yongxin Tong, Jindong Han, Panpan Zhang, Xinjiang Lu, and Hui Xiong. 2020. Incorporating multi-source urban data for personalized and context-aware multi-modal transportation recommendation. *IEEE Transactions on Knowledge and Data Engineering* (2020).
[19] Qiang Liu, Shu Wu, Liang Wang, and Tieniu Tan. 2016. Predicting the next location: A recurrent model with spatial and temporal contexts. In *Thirtieth AAAI conference on artificial intelligence*.
[20] Ruiqiang Liu, Shuai Zhao, Bo Cheng, Hao Yang, Haina Tang, and Fangfang Yang. 2020. ST-MFM: A Spatiotemporal Multi-Modal Fusion Model for Urban Anomalies Prediction. In *ECAI 2020*. IOS Press, 1922–1929.
[21] Sara Proença and Elias Soukiazis. 2008. Tourism as an economic growth factor: a case study for Southern European countries. *Tourism Economics* 14, 4 (2008), 791–806.
[22] Honglei Ren, You Song, Jingwen Wang, Yucheng Hu, and Jinzhi Lei. 2018. A deep learning approach to the citywide traffic accident risk prediction. In *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 3346–3351.
[23] Ankur Sarker, Haiying Shen, and Kamran Kowsari. 2020. A Data-Driven Reinforcement Learning Based Multi-Objective Route Recommendation System. In *2020 IEEE 17th International Conference on Mobile Ad Hoc and Sensor Systems (MASS)*. IEEE, 103–111.
[24] Panote Siriaraya, Yuanyuan Wang, Yihong Zhang, Shoko Wakamiya, Péter Jeszenszky, Yukiko Kawai, and Adam Jatowt. 2020. Beyond the Shortest Route: A Survey on Quality-Aware Route Navigation for Pedestrians. *IEEE Access* 8 (2020), 135569–135590.
[25] Joanna Szlapczynska. 2015. Multi-objective weather routing with customised criteria and constraints. *The Journal of Navigation* 68, 2 (2015), 338–354.
[26] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *arXiv preprint arXiv:1706.03762* (2017).
[27] Junshan Wang and Guojie Song. 2018. A deep spatial-temporal ensemble model for air quality prediction. *Neurocomputing* 314 (2018), 198–206.
[28] Xiaoyang Wang, Yao Ma, Yiqi Wang, Wei Jin, Xin Wang, Jiliang Tang, Caiyan Jia, and Jian Yu. 2020. Traffic flow prediction via spatial temporal graph neural network. In *Proceedings of The Web Conference 2020*. 1082–1092.
[29] Mingxing Xu, Wenrui Dai, Chunmiao Liu, Xing Gao, Weiyao Lin, Guo-Jun Qi, and Hongkai Xiong. 2020. Spatial-temporal transformer networks for traffic flow forecasting. *arXiv preprint arXiv:2001.02908* (2020).
[30] Bin Yang, Chenjuan Guo, Yu Ma, and Christian S Jensen. 2015. Toward personalized, context-aware routing. *The VLDB Journal* 24, 2 (2015), 297–318.
[31] Binbin Zhou, Longbiao Chen, Sha Zhao, Fangxun Zhou, Shijian Li, and Gang Pan. 2021. Spatio-temporal analysis of urban crime leveraging multisource crowdsensed data. *Personal and Ubiquitous Computing* (2021), 1–14.
[32] Binbin Zhou, Longbiao Chen, Fangxun Zhou, Shijian Li, Sha Zhao, Sajal K Das, and Gang Pan. 2020. Escort: Fine-grained urban crime risk inference leveraging heterogeneous open data. *IEEE Systems Journal* (2020).
[33] Zhengyang Zhou, Yang Wang, Xike Xie, Lianliang Chen, and Hengchang Liu. 2020. RiskOracle: A Minute-Level Citywide Traffic Accident Forecasting Framework. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34. 1258–1265.